Unlearning in GANs via Recursive Post-training

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Abstract

Machine unlearning aims to modify a trained model to have it perform poorly on a select subset of the data. This is an increasingly popular task with as applications such as avoiding copyright infringement or censoring inappropriate content. We focus on unlearning in the context of image generation, in particular targeting generative adversarial networks (GANs).

In this work, we propose a simple yet effective unlearning procedure for GANs that requires access solely to a pretrained generator and a small forget set. This procedure leverages the catastrophic forgetting tendencies of GANs in a controlled manner, ensuring that the generator naturally stops generating images associated with the forget set while largely preserving image quality on the rest of the data. Our results demonstrate that targeted unlearning can be achieved with significantly fewer training epochs than simply retraining from scratch and with minimal degradation in image quality.

1 Introduction

As machine learning systems are increasingly used in real-world settings, the ability to selectively remove the influence of specific training data from these ever-learning models has become more desired. With the vast sources of data that large models such as LLMs use to train, it is nearly impossible to ensure that all of this training data matches the requirements set by companies and users. Some examples of undesired data includes harmful content like nudity and data taken from disallowed sources such as copyrighted or private data. These requirements have led to the growing field of machine unlearning, which aims to remove the effect of a designated subset of training data without retraining a model from scratch while preserving performance on the remaining data.

Most existing unlearning methods have focused on classification tasks. In this setting, explicit loss functions make it easier to give the model an objective to unlearn certain features of the data. By comparison, relatively little work has looked at unlearning in the context of generative adversarial networks (GANs). GANs have a very specific training routine and loss which doesn't allow many of the techniques used in classification models to be easily applied.

Full retraining without the targeted forget set is the most reliable approach to removing unwanted data from GANs but has high computational cost. In many realistic scenarios, retraining is impractical, particularly when access to the original training data is limited or when the training data is too large. In this work, we study the problem of unlearning in GANs when only a pretrained generator and a small forget set are available. Rather than attempting to precisely undo the contribution of individual samples through cost functions, we leverage a defining property of GANs: their susceptibility to catastrophic forgetting. Typically this is a failure mode for GANs, but we can use this

behavior as a mechanism for targeted unlearning. By filtering out generated samples that resemble the forget set and doing post-training on the remaining data, we encourage the model to suppress undesired features while preserving overall image quality.

2 Related work

2.1 Generative Adversarial Networks

Generative Adversarial Networks (GANs) [2] are a popular technique of image generation models. This technique involves two models, the generator and the discriminator, trained simultaneously in an adversarial manner. The generator takes in random Gaussian noise and outputs generated images, while the discriminator learns to classify whether a given image comes from the generator or from the real data.

The discriminator loss is that of standard prediction:

$$\mathcal{L}_D = \mathbb{E}_{x\sim X} \log p_D(x) + \mathbb{E}_{x\sim G(z), z\sim N(0,I)}(1 - \log p_D(x))$$

where X is the true data.

The generator has the loss of fooling the discriminator:

$$\mathcal{L}_D = \mathbb{E}_{x\sim G(z), z\sim N(0,I)}(\log p_D(x))$$

This creates a min-max game whose optimum is the generation of realistic images:

$$\min_G \max_D \mathbb{E}_{x\sim X}[\log p_D(x)] + \mathbb{E}_{x\sim G(z), z\sim N(0,I)}[1 - \log p_D(x)]$$

2.2 Unlearning in classification tasks

Much of the research in unlearning has focused on classification models. In these models, given some underlying forget distribution of all the data we would like to unlearn, the goal is to have the model perform similarly to a model that was trained without access to this forget distribution. Formally, there is some underlying data distribution D, forget distribution $D_f \subset D$, and retain distribution $D_r = D \setminus D_f$. We are given pretrained model $\theta := M(D)$ trained on D, and we want to train $\hat{\theta}$ such that:

$$\forall d \in D, p_{\hat{\theta}}(x) \approx p_{\theta_r}(x)$$

where $\theta_r = M(D_r)$ is the model only trained on the retain distribution.

The most rudimentary technique for achieving this unlearning is gradient ascent, doing post-training where the loss of the model on datapoints in the forget set is maximized rather than minimized. This technique, however, is known to greatly hinder the performance of the model on the retain set[5]. Several similar alternative approaches have been tried to mitigate this. NegGrad+ [5] tries to minimize the loss on the retain set while simultaneously maximizing the loss on the forget set. The same authors also propose SCRUB, where instead of doing gradient descent/ascent with the true labels, they do gradient descent/ascent using the labels that the pretrained model output. SalUn [1] performs gradient descent with random labels assigned to data in the forget set, and additionally only update certain parameters computed to be highly salient to the desired unlearn feature. Tarun et al. [10] train with forget set data and random noise labels on a high learning rate to unlearn the forget distribution quickly, and then retrain on the retain set to repair the damage. Warnecke et al. [13] explore the use of influence functions for unlearning, which measure the effect of upweighting a certain datapoint on the parameters of the model. By estimating the effect of upweighting a datapoint on the parameters of the model, they are able to estimate what the parameters would look like if the datapoint were downweighted to 0 weight, ie, if it were removed from the dataset.

2.3 Unlearning in GANs

For the task of image generation, it would be tempting to use the same objective, to have the model perform similarly as it would if it were trained on only the retain set. However, because the objective of image generation models is to exactly match the distribution of images in the underlying data, we can instead give it the objective to generate images in the forget distribution with probability 0, ie:

$$\forall x \in D_f, p_{\hat{\theta}}(x) = 0$$

This is what Kong and Chaudhuri [4] denote image redaction, but we stick with unlearning as that is the standard terminology used in the literature. There has been some research on adopting techniques similar to those of classification to image generation by training image-to-image autoencoder models with additional metrics to encourage poor accuracy on the forget set. When applied to GANs, this means using GAN *inversion*, the training of a model to invert the GAN's output $\hat{G}^{-1}(G(z)) \approx z$. Li et al. [6] for example incorporate the objective of reconstructing forget set images as random noise. Sun et al.[9] and Moon et al.[7] incorporate the objective of reconstructing forget set images as the average image.
In this paper, we focus instead on a different approach, that of leveraging catastrophic forgetting. Catastrophic forgetting is the tendency of a model to perform poorly on data that it has not encountered in its training for some time. This is normally a flaw in a model, but we can leverage it to perform unlearning by continuing to train on the retain set, hoping that the model will stop generating the forget set because it does not encounter it during this post-training.

Prior research has shown that GANs are particularly susceptible to catastrophic forgetting [11] [14] which makes GANs a promising target for this approach.
This approach has been explored before, but the exact setup in prior research is different than what we are pursuing here. [4] assume that we have access either to the original retain-set data or that we have access to a ground truth classifier. Tiwary et al. [12] similarly assume access to a ground truth classifier, and also implement a more complex procedure in which they add an extra term penalizing the parameters of the model for being too similar to one generating forget set images. In constrast, in this paper, we assume only access to a forget set.

3 Methods

3.1 Problem formulation

We are given model $\theta := M_t(D)$ trained on samples from the underlying data distribution D. We have some forget distribution $D_f \subset D$ we would like to unlearn, and are given samples from it that we call the forget set:

$$F \sim D_f$$

Our goal is to train model $\hat{\theta} := M_u(\theta, F)$ such that the model does not generate from the forget set:

$$\forall x \in D_f, p_{\hat{\theta}}(x) = 0$$

and outside of the forget set, the distribution stays the same:

$$D_r = D \setminus D_f$$

$$\forall x \in D_r, p_{\hat{\theta}}(x) = \frac{p_\theta(x)}{p_D(D_r)}$$

3.2 Pipeline

Our primary contribution is the following pipeline.
First, we use samples from the forget set to train a classifier $C : D_\theta \to \{0, 1\}$ to predict if a given image comes from the forget set, treating it as an outlier detection problem. We then take a pretrained generator $\theta := M(D)$ and generate a large number of images $\{\hat{x}\}_1^N \sim D_\theta$. Note that we cannot simply train the classifier in a standard supervised fashion by treating the dataset generated by the pretrained model as retain set images because some of these images are actually forget images.

Next we use the classifier to filter out forget images from the generated set and treat the rest of them as the new ground truth set: $Y := \{\hat{x} : C(\hat{x}) = 0\}$. Finally we continue to train θ in standard GAN fashion on this new ground truth set with a brand new discriminator:

$$\mathcal{L}_G = \mathbb{E}_{x \sim D_\theta}[\log p_D(x)]$$

$$\mathcal{L}_D = \mathbb{E}_{x \sim D_\theta}[1 - \log p_D(x)] + E_{x \sim Y}[\log p_D(x)]$$

3.3 Classifier

Classifying data using only positive examples of a class is challenging because the model has no direct information about what lies outside that class. This one-class classification problem is often addressed by finding the smallest hypersphere that encloses all training samples in some latent representation of the images, and treating every datapoint outside that hypersphere as an outlier. Formally, the problem can be written as

$$\min_{r,\,c} \; r^2 \quad \text{subject to} \quad \|\phi(x_i) - c\|^2 \leq r^2, \qquad i = 1, \ldots, n,$$

where c denotes the center of the hypersphere, r its radius, and $\phi(\cdot)$ a mapping into feature space. In practice, enforcing the above hard constraints can lead to degenerate solutions where our hypersphere collapses. To mitigate this, we use the following problem instead

$$\min_{r,\,c} \; r^2 + \frac{1}{\nu n} \sum_{i=1}^{n} \max\left(0,\; \|\phi(x_i) - c\|^2 - r^2\right).$$

Where ν controls the trade off between encouraging a small radius and permitting boundary violations, and the hinge penalty allows some points to remain outside of our boundary.

We train an encoder network ϕ using this loss with the goal of making the representations of the forget set samples as close as possible to the hypersphere center c. At test time, samples that fall outside the learned hypersphere are considered anomalies and thus not part of the forget set.

4 Experimental Details

4.1 Datasets

We experiment on two datasets: handwritten digits (MNIST) and facial images (CelebA-HQ). MNIST contains 70,000 handwritten digits 28x28 grayscale images roughly equally distributed among the ten digits, providing a good toy proof-of-concept dataset for which we can conduct thorough testing. We select images of the digit 1 to be our forget set. CelebA-HQ contains 30,000 aligned and cropped celebrity face images at 256×256 resolution, annotated with 40 attributes. We choose images with the Eyeglasses attribute to be our forget set, which comprise roughly 6.51% of the dataset.x

4.2 Models

For the MNIST experiments, we use a pretrained Deep Convolutional GAN (DCGAN) trained for 100 epochs on the full dataset. The generator takes a 100-dimensional latent vector and creates 28×28 grayscale images using four ConvTranspose layers with BatchNorm and ReLU activations. The discriminator mirrors this structure with four convolutional layers and LeakyReLU activations to classify real from generated digits.

For the facial image experiments, we use a Progressive GAN (PGAN) loaded from the `facebookresearch/pytorch_GAN_zoo` repository, pretrained on the celebAHQ-256 task. The generator maps a 512-dimensional latent vector to 256×256 RGB images, and the discriminator follows the standard PGAN architecture used during pretraining on CelebA-HQ.

SVDD classifiers are adapted to each domain. For MNIST, the SVDD uses an standard feed-forward encoder trained exclusively on digit 1. For CelebA-HQ, the SVDD employs a three-layer CNN encoder applied separately to each RGB channel of 64×64 images, projecting them into a 32-dimensional embedding space in which the Eyeglasses attribute defines the center of the hypersphere. A small note is that GAN-generated images often contain subtle noise patterns that aren't detected by the human eye. To mitigate the impact of this noise on the encoder, we uniformly add small random noise around the image before feeding it into the network. In practice, we have found that this helps the hypersphere better encompass points that do not correspond to real data, improving the robustness of our model.

FID score	post-train from pretrained model	Retrain from scratch
Artificial data	**51**	117
Real data	51	50

Table 1: Comparison of Image Quality (FID Score)

4.3 Hyperparameters

For MNIST, the generator (G) and discriminator (D) are optimized using Adam with learning rate of 2×10^{-4} and betas $= (0.5, 0.99)$. We give the discriminator a warm start of 20 epochs before training the generator. The generator is updated twice between every update of the discriminator. We train with a batch size of 1024 for 50 epochs for post-training, and 100 epochs for retraining. For the classifier, we add gaussian noise with mean=0 and standard deviation 0.1 to all input images.

For CelebA-HQ, the generator (G) and discriminator (D) are trained for 100 epochs with a batch size of 16. The learning rate for both G and D is set to 5×10^{-5}, which is lower than standard training to preserve the pretrained model quality. The discriminator is updated 5 times for every single generator update. Additionally, an L1 loss is applied between the unlearning model and the frozen original model to preserve the generator's output and reduce noise. Both networks are optimized using Adam with betas $= (0.0, 0.99)$.

5 Results

We evaluate the effectiveness of our pipeline along two dimensions. To measure image quality, we use the Frechet Inception Distance (FID)[3] between generated images and the true retain set. To measure unlearning effectiveness, we measure the proportion of generated images classified as belonging to the forget set. For this classification, we train a separate ground-truth classifier on the full dataset.

We include three models as baselines. First, we retrain a model from scratch using our artificial dataset, illustrating the contribution of our artificial dataset to the results. Second, we do the same post-training but assuming that we have access to the original dataset and the true classification of the forget set in that dataset. Third, we combine the two baselines, retraining from scratch assuming access to the dataset, illustrating a lower bound for what we could hope to achieve by using a GAN. Alongside our approach, these four models make up a 2x2 grid shown in our tables

5.1 MNIST

The pretrained DCGAN baseline exhibited an FID score of 52, with the target forget class (digit 1) being approximately 10% of the generated output, consistent with the uniform distribution of the dataset.

As shown in Table 1, our approach post-training artificial data successfully preserves image quality, achieving an FID of 51. This significantly outperforms retraining from scratch on artificial data, which suffers from severe image degradation (FID 117).

When retraining from scratch using real data, an ideal scenario unavailable in our problem setting, the model improves only marginally to an FID of 50. This close performance validates our approach, showing that it introduces minimal noise compared to the theoretical optimum.

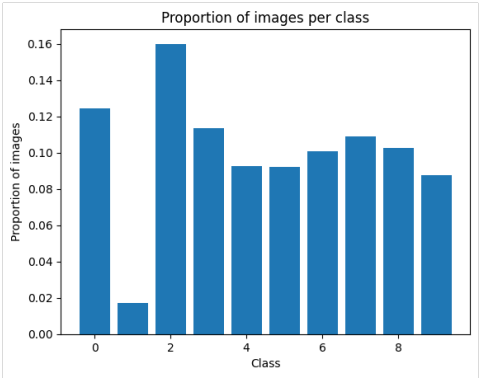
Table 2 illustrates the effectiveness of our approach in reducing the generation of the target forget class. Our post-training method lowers its prevalence from the baseline 10% to 2.375%. Although retraining from scratch on artificial data achieves a lower rate of 0.81%, this comes at the cost of degraded image quality (FID 117). This suggests that the baseline reduces the target class primarily through image degradation rather than targeted unlearning.

% ones	post-train from pretrained data	Retrain from scratch
Artificial data	**2.375%**	0.81%
Real data	0.75%	0.125%

Table 2: Unlearning Effectiveness (% Ones Generated)



(a) Digit distribution in pretrained model

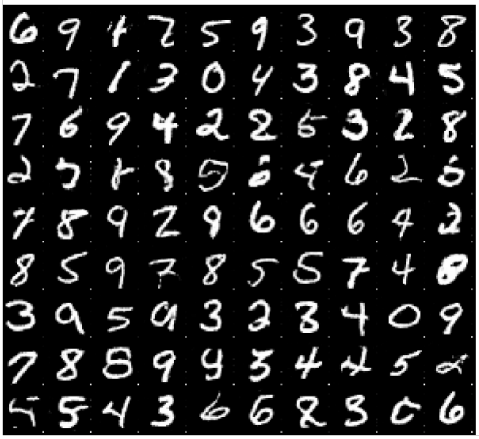


(b) Digit distribution in model after training

Figure 1: Comparing the distribution over digits before and after post-training



(a) Pretrained gan output

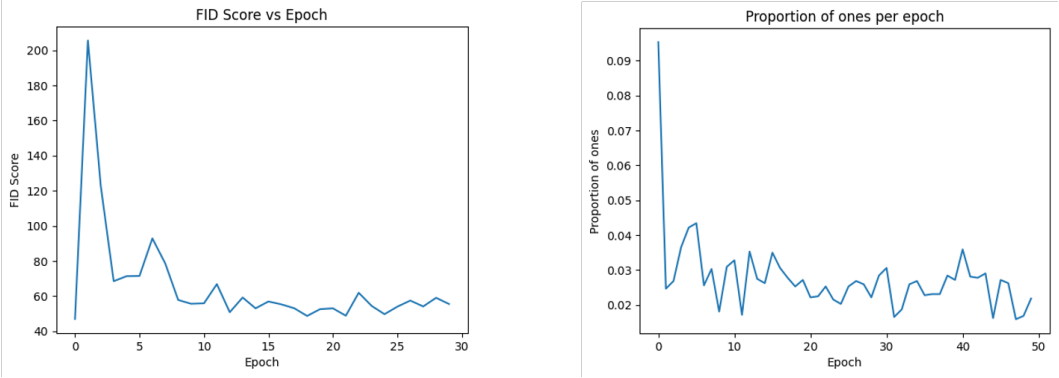


(b) post-trained gan output

Figure 2: Comparing images generated before and after post-training

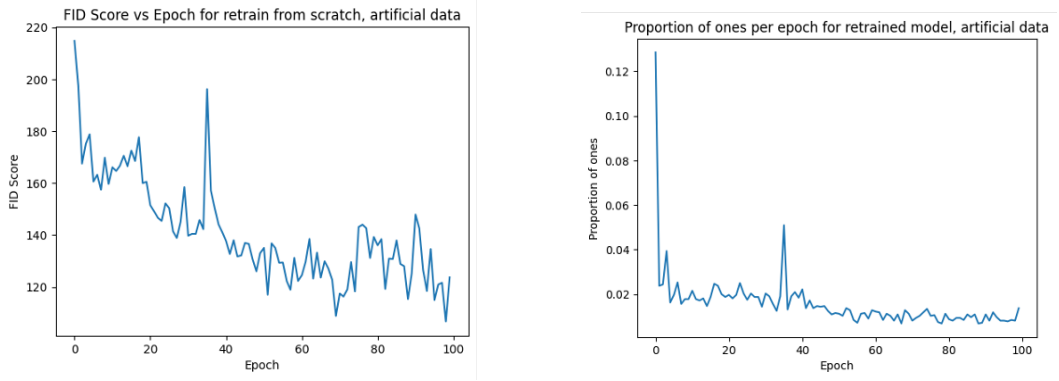
In Figure 1, we also display the distribution of the rest of the digits after training. As can be seen, the distribution of the other digits is roughly uniform. The percentage of twos is slightly higher than would be seen in a uniform distribution, but we attribute this to the pretrained model also having a slightly higher proportion of twos.

To show the increased computational efficiency in our unlearning method in comparison to retraining from scratch, we also display the graph of the FID score and percentage ones over time while training in figures 3 and 4. As can be seen, it takes around 15 epochs for the post-training to max out, vs. around 70 epochs when retraining from scratch. This confirms that our approach is much more computationally efficient than retraining from scratch.



(a) FID score vs epoch (b) Ones per epoch

Figure 3: post-training pipeline



(a) FID score per epoch (b) Ones per epoch

Figure 4: Retraining benchmark using artificial data

5.2 CelebA-HQ

Due to computational constraints, our evaluation on the CelebA-HQ dataset was limited to a single experiment tracking the prevalence of the forget class (eyeglasses) throughout post-training. Unlike the MNIST setting, we were unable to run full baselines or compute image-quality metrics.

As shown in Figure 5, the probability that the generator produces images containing eyeglasses exhibits a consistent downward trajectory over 100 epochs. This suggests that the post-training pipeline can suppress the targeted attribute even in a complex, high-dimensional latent space.

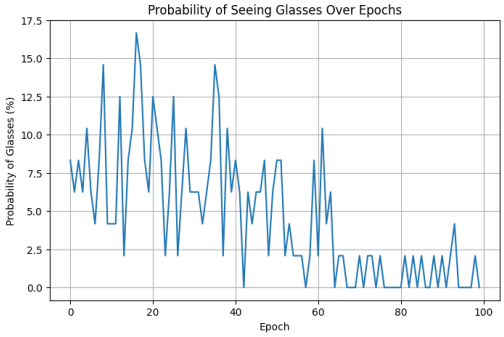


Figure 5: Eyeglasses per epoch

6 Discussion

Based on our results, retraining a model from scratch on real data remains the strongest baseline. However, this approach is often infeasible in practice, as access to the original dataset is often restricted and retraining from scratch is computationally expensive. Under these constraints, our recursive post-training strategy offers a practical alternative.

A key observation from our experiments is the contrast between our approach and the artificial-data retraining baseline. As shown in Table 1, models trained from scratch using only generated samples produced highly degraded images (FID 117). This occurs because the generator must relearn the full distribution from noisy artificial data. In contrast, our method starts from a pretrained model that already captures the global data manifold. The recursive training loop needs only to remove a specific subset of this distribution rather than reconstruct it entirely.

A limitation of our approach is that it still retains a small 2.375% of the forget class, compared to 0.125% for the real-data baseline. The unlearning process depends on the accuracy of the SVDD classifier, since any forget-set samples misclassified as clean are reinforced by the generator in subsequent iterations, representing a key constraint of the method.

Another important limitation of our method is that the class of forget images must be learnable so that the classifier can correctly filter out what generated images are part of the forget distribution. For example, our method likely wouldn't work well on copyrighted images because a classifier would not be able to model well what makes an image copyrighted.

Our procedure does not necessitate that the pretrained model is originally trained as a GAN. Future research could experiment with applying this technique to models originally trained with another image generation architecture such as the popular latent diffusion models [8].

7 Conclusion

We introduce a straightforward pipeline for unlearning classes of images in GANs. This pipeline assumes access only to a forget set and to a pretrained model and needs no access to the original dataset. The pipeline works by leveraging the tendency of GANs to exhibit catastrophic forgetting. Our experiments on MNIST and CelebA-HQ show that our model is able to effectively reduce the number of unlearned images in the dataset in far fewer iterations than retraining from scratch and with comparable or even better image quality.

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